

Lecture 15

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Constrained optimization

The most common optimization problems come with constraints; the additional requirements on a program can make a big difference in the **algorithm choice**, **runtime**, and **solvability** of a problem.

We illustrate the distinction in the following two examples.

Example (Unconstrained quadratic programming)

Let $A \in \mathbb{S}^n$, $\mathbf{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$. In the last homework, we show that

$$\min_{\mathbf{x} \in \mathbb{R}^n} \frac{1}{2} \mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$$

has a unique solution iff $A \succeq 0$, and that a closed-form solution was available in that case.

Example (Linearly constrained quadratic programming)

Let $A, \mathbf{b}, c$ be as above, and let $\mathbf{h} \in \mathbb{R}^m$ and $G \in \mathbb{R}^{m \times n}$. The problem

$$\begin{aligned} & \text{minimize } \frac{1}{2} \mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c \\ & \text{subject to } G \mathbf{x} \leq \mathbf{h} \end{aligned}$$

This problem is NP-hard unless A is positive semidefinite.

An example of an NP-hard problem that can be written in this form is the Facebook friendship clique problem.

Friendship clique problem

The idea is that we have an adjacency matrix A which captures the friendship relationships between users:

$$A_{ij} = 1 \iff \text{user } i \text{ is friends with user } j; \quad A_{ij} = 0 \text{ otherwise}$$

We want to find the largest “clique” of users who are all friends with each other.

Let $x_i = 1$ if user i is in the clique, and $x_i = 0$ otherwise.

$$\begin{aligned} & \text{minimize } - \sum_{i=1}^n x_i \quad (\text{equivalent to maximizing } \sum_{i=1}^n x_i) \\ & \text{subject to } x_i + x_j \leq 1 \quad \text{for all } (i, j) \text{ where } A_{ij} = 0 \\ & \quad \quad \quad 0 \leq x_i \leq 1 \quad \text{for all } i \end{aligned}$$

To put it exactly in the form $\frac{1}{2} \mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$:

$$\begin{aligned} & \text{minimize } \frac{1}{2} (0) \mathbf{x}^\top \mathbf{x} + (-\mathbf{1})^\top \mathbf{x} + 0 \\ & \text{subject to } x_i + x_j \leq 1 \quad \text{for all } (i, j) \text{ where } A_{ij} = 0 \\ & \quad \quad \quad 0 \leq x_i \leq 1 \quad \text{for all } i \end{aligned}$$

To get the constraints in the form $G\mathbf{x} \leq \mathbf{h}$, we need to define a matrix M where each row corresponds to a non-edge in the graph (i.e., where $A_{ij} = 0$), and that row has 1's in positions i and j and 0's elsewhere. Then we can write the problem as:

$$\begin{aligned} & \text{minimize} && \frac{1}{2}(\mathbf{0})\mathbf{x}^\top\mathbf{x} + (-\mathbf{1})^\top\mathbf{x} + 0 \\ & \text{subject to} && \begin{bmatrix} I \\ -I \\ M \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{1} \end{bmatrix} \end{aligned}$$

The constraints $I\mathbf{x} \leq \mathbf{1}$ and $-I\mathbf{x} \leq \mathbf{0}$ together enforce $0 \leq x_i \leq 1$

The constraint $M\mathbf{x} \leq \mathbf{1}$ enforces that no two non-adjacent vertices can both be in the clique

If A is positive semidefinite, then the problem can be solved in polynomial time, though there is no closed-form solution.

Another important special case is when $A = 0$ and which case we obtain linear programming or LP

$$\begin{array}{ll}\text{minimize} & \mathbf{b}^\top \mathbf{x} \\ \text{subject to} & G\mathbf{x} \leq \mathbf{h}\end{array}$$

Although constrained optimization appears easier to solve because of the smaller domain, the above examples illustrate the difficulties associated with adding constraints.

We now formally introduce equality constrained optimization and inequality constrained optimization problems.

Equality and inequality constrained optimization

Definition (Equality Constrained problem)

A problem of the form

$$\begin{aligned} & \text{minimize } f(\mathbf{x}) \\ & \text{subject to } g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m \end{aligned}$$

is called an **equality constrained** optimization problem.

Definition (Inequality Constrained problem)

A problem of the form

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ & g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m \\ \text{subject to} & h_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, p\end{array}$$

is called an **inequality constrained** optimization problem.

Note that

$$g_i(\mathbf{x}) = 0 \quad \Leftrightarrow \quad -g_i(\mathbf{x}) \leq 0 \text{ and } g_i(\mathbf{x}) \leq 0$$

so every constrained problem can be written as an inequality constrained problem.

In most cases, $f, g_i, h_i : \Omega \rightarrow \mathbb{R}$ where $\Omega \subset \mathbb{R}^n, \mathbb{S}^n$, or $\mathbb{R}^{m \times n}$ is an open set.

Lagrange multipliers

How do we deal with constraints? One approach is the method of **Lagrange multipliers**.

The basic idea is to convert the constrained problem into an unconstrained one by incorporating the constraints directly into the objective function.

Define $G : \Omega \rightarrow \mathbb{R}^m$ by

$$G(\mathbf{x}) = \begin{bmatrix} g_1(\mathbf{x}) \\ \vdots \\ g_m(\mathbf{x}) \end{bmatrix}$$

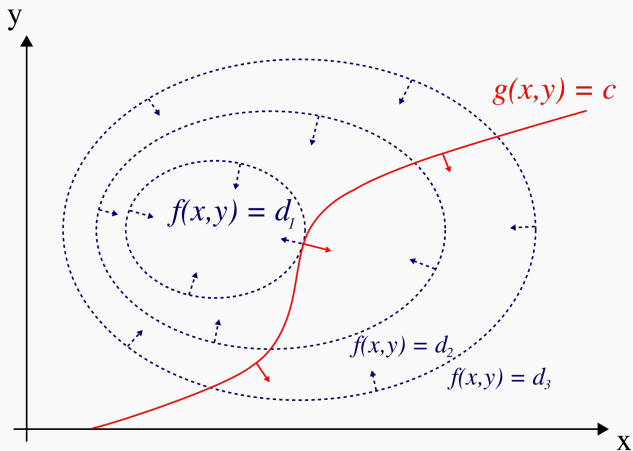
Then the equality constrained optimization problem can be written as

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{subject to} & G(\mathbf{x}) = \mathbf{0} \end{array}$$

We assume $m \leq n$ or otherwise we may have $\{\mathbf{x} \in \Omega : G(\mathbf{x}) = \mathbf{0}\} = \emptyset$ (overdetermined systems of equations).

Optimality conditions

Just like the case of unconstrained optimization, we are going to need optimality conditions in order to design algorithms.



(Image By Nexcis <https://commons.wikimedia.org/w/index.php?curid=3831272>)

A potential optimizing point is one where the constraint is satisfied and where moving in any direction that still satisfies the constraint will not result in a change in the objective function. In other words, moving in any direction that satisfies the constraint will keep the objective function constant.

The gradient of the constraint is perpendicular to any movement direction that will keep the constraint satisfied.

The gradient of the objective function is perpendicular to any movement direction that will keep the objective function constant.

Since we want to find points where these two movement directions are parallel, we need to find points where the gradient of the objective function is a linear combination of the gradients of the constraints.

Theorem (Necessary condition for equality constrained optimality)

Let $f, g_1, \dots, g_m \in C^1(\Omega)$. If $\mathbf{x}_*$ is a local minimizer of $(\star)$ and $JG(\mathbf{x}_*) \in \mathbb{R}^{m \times n}$ has rank m , then there exist $\lambda_1^*, \dots, \lambda_m^*$ such that

$$\nabla f(\mathbf{x}_*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}_*) = \mathbf{0}$$

or, alternatively

$$\nabla f(\mathbf{x}_*) + \lambda^*{}^T JG(\mathbf{x}_*) = \mathbf{0}$$

where $\lambda^* = [\lambda_1^*, \dots, \lambda_m^*]^T$.

Proof. Implicit function theorem.

The corresponding result for general constrained optimality with both equality and inequality constraints is more complicated and is known as the **Karush-Kuhn-Tucker** or KKT conditions, by far the most important conditions in optimization theorem.

Theorem (Karush-Kuhn-Tucker conditions)

Let $\mathbf{x}_*$ be a local minimizer of (**):

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ & g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m \\ \text{subject to} & \\ & h_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, p\end{array}$$

There exist $\lambda_1^*, \dots, \lambda_m^*, \nu_1^*, \dots, \nu_p^* \in \mathbb{R}$ such that satisfy **stationarity**, **primal feasibility**, **dual feasibility**, and **complementary slackness**.

Stationarity condition (same as the necessary condition for equality constrained optimality):

$$\nabla f(\mathbf{x}_*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}_*) + \sum_{i=1}^p \nu_i^* \nabla h_i(\mathbf{x}_*) = \mathbf{0},$$

Primal feasibility: We are at a point where the constraints are satisfied.

$$\begin{aligned} g_i(\mathbf{x}_*) &= 0, & i &= 1, \dots, m, \\ h_i(\mathbf{x}_*) &\leq 0, & i &= 1, \dots, p, \end{aligned}$$

Dual feasibility: We are at a point where the Lagrange multipliers for the inequality constraints are non-negative.

$$\nu_i^* \geq 0, \quad i = 1, \dots, p,$$

Complementary slackness: If an inequality constraint is not active (that is, we're not pushing up against the boundary of where the constraint is satisfied), the corresponding Lagrange multiplier must be zero.

$$\nu_i^* h_i(\mathbf{x}_*) = 0, \quad i = 1, \dots, p.$$

William Karush first established the theorem as part of his MSc in Mathematics thesis at the University of Chicago. Harold Kuhn and Albert Tucker discovered the conditions independently of Karush almost twelve years later but received name recognition because they were already famous. Tucker was a postdoc here.

Definition

The function defined by

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top G(\mathbf{x}) + \boldsymbol{\nu}^\top H(\mathbf{x}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x})$$

is called the **Lagrangian** of the optimization problem (**).

The variables $\mathbf{x} = (x_1, \dots, x_n)$ are called **primal variables**.

The variables

$$\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_m)$$

$$\boldsymbol{\nu} = (\nu_1, \dots, \nu_p)$$

are called **dual variables**.

Note that the stationarity condition in the KKT conditions is nothing more than $\nabla L(\mathbf{x}_*, *, *) = \mathbf{0}$.